

PAPER_1355 — Jamming Transition ϕ_J

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED) **Calculator:**
`calculate_paradox({"paradox": "jamming_transition"})`

Master Expression

$$\phi_J = 2/3 = (D_{\text{phys}}-1)/D_{\text{phys}} = 0.667$$

Match: 4% from obs 0.64

Interpretation

Random close packing density at $(D_{\text{phys}}-1)/D_{\text{phys}}$ geometric threshold.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$
- $\omega_{\text{SCm}} = 1.25$ THz

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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